Muhammad Sarmad Sohail

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Education

National University of Sciences and Technology (NUST) *Islamabad, Pakistan*
Bachelor of Electrical Engineering *Sep. 2020 – Jun. 2024*

- **CGPA:** 3.70/4.00
- **Thesis:** Resource Allocation for MEC-Enabled CR-NOMA Networks using DRL | Paper | Demo | Poster
  Advised by Dr. Syed Ali Hassan
- **Key Courses:** Mobile Communication Systems, Machine Learning, Digital Signal Processing, Embedded System Design, Robotics, Digital System Design, Bitcoin & Cryptocurrencies, Probability & Statistics

Industry Experience

Adept Tech Solutions (ATS) *Sept. 2025 – Present*
Data Scientist *Pleasanton, California (Remote)*

- Develop and maintain production-grade modules for the ASCEND Synthetic Data Platform: statistical analysis, preprocessing, data profiling, feature engineering, anonymization, curation, and synthetic data generation, exposed as **Python** and **FastAPI** microservices
- Implement these modules as distributed **PySpark** jobs and orchestrate ETL and generation workloads for scalable execution across large datasets
- Build the platform's **rule-based generation engine**, layering configurable business logic over statistical generation with composite foreign-key resolution and per-parent cardinality for referentially correct data
- Build and evaluate tabular **synthetic data generators** across three model families: **GAN**, **autoregressive**, and **cVAE**, for fraud detection on financial, insurance, and medical datasets under severe class imbalance
- Raised synthetic fidelity on a key transaction behaviour to **98.78%** (98.96% for real data) by correcting the training feature set, with generators trained and tuned on **AWS GPU** instances

Xcelerium *Jan. 2025 – May. 2025*
Trainee Engineer - Analog & Mixed Signal IC Design *Islamabad, Pakistan*

- Designed a 100 MHz 7-bit SAR ADC in **TSMC 28nm** PDK using **Cadence Virtuoso**, integrating bootstrapped sampling switches, dynamic latch comparators, and a binary-weighted capacitive DAC
- Designed a high-speed 12-bit SAR ADC prototype in **GF 22nm FD-SOI** PDK using **Synopsys Custom Compiler**, implementing triple-tail comparators, non-overlapping clock generation, and custom SAR logic blocks (targeting 1GHz)
- Streamlined **AMS design workflow** by formulating a schematic-to-layout toolflow plan, improving standardization in Synopsys environments

NUST Chip Design Centre (NCDC) *Jul. 2024 – Nov. 2024*
Analog & Mixed Signal IC Design Trainee *Islamabad, Pakistan*

- Hands-on training in MOS devices, analog design cycles, schematic and layout design using **Cadence Virtuoso**
- Practical experience in designing OTAs, BGR, LDO, VCO, PLLs, and performing **DRC, LVS,** and chip sign-off processes
- Collaborated on the design of a **Passive RFID Transponder Chip** for animal tagging using **CMOS 65nm** technology

Research Experience

Pak Angels – Generative AI Training Program | GitHub *Jun. 2025 – Jul. 2025*
GenAI Trainee *Remote, Pakistan*

- Completed intensive training on **Generative AI**, ChatGPT APIs, Python, **LangChain**, **RAG systems**, and autonomous AI agents
- Delivered weekly hands-on projects and participated in final hackathon; awarded Top Performer among cohort

Information Processing & Transmission Lab, SEECS | GitHub *May 2023 – May 2024*
Research Intern *Islamabad, Pakistan*

- Designed and implemented a **Deep Deterministic Policy Gradient (DDPG)** framework integrating **Mobile Edge Computing (MEC)** and **CR-NOMA** for dynamic resource allocation in energy-harvesting IoT networks
- Published research at **IEEE WCNC 2024**, demonstrating faster convergence and superior performance over baseline offloading methods (random, always, never offloading)
- **Tech Stack:** Python, TensorFlow, Deep RL (DDPG), Simulation & Analysis

Optimization & Machine Learning Lab, SEECS | GitHub — *Jun. 2023 – Aug. 2023*
Research Intern — *Islamabad, Pakistan*

- Developed novel **Conv2D + LSTM** architectures for video-based regression tasks on the **UBFC dataset**
- Preprocessed video datasets and implemented data augmentation pipelines for training deep CNN models
- **Tech Stack:** Python, PyTorch, Keras, NumPy, Matplotlib

Publications

Optimizing Resource Allocation in MEC-Enabled CR-NOMA-Assisted IoT Networks: A DRL-Driven Strategy — *May 2024*
IEEE Wireless Communications & Networking Conference (WCNC) 2024 | Dubai, UAE
Muhammad Taha Qaiser, Muhammad Sarmad Sohail, Minahil Shafqat, Syed Asad Ullah, Haejoon Jung, Syed Ali Hassan

Key Projects

Passive RFID Transponder Chip Design — *Aug. 2024 – Nov. 2024*
NUST Chip Design Centre | Cadence Virtuoso, CMOS 65nm — *Islamabad, Pakistan*

- Designing RFID transponder chip for animal tagging, focusing on ultra-low power consumption, reliability, and market competitiveness

FYP: Optimizing Resource Allocation in NOMA Networks — *May 2023 – May 2024*
IPT Lab, SEECS | Python, TensorFlow, DDPG — *Islamabad, Pakistan*

- Developed a **Deep Deterministic Policy Gradient (DDPG)** agent for MEC-enabled CR-NOMA networks, dynamically optimizing time-sharing between data transmission and energy harvesting for resource-constrained IoT devices
- Achieved minimized service delay for computation offloading through extensive simulation, outperforming baseline methods with faster convergence and robust performance

4-bit Microprocessor and VGA Line Drawing — *Apr. 2023 – Jun. 2023*
SEECS, NUST | Verilog, FPGA — *Islamabad, Pakistan*

- Designed a custom 4-bit microprocessor with hierarchical datapath and control logic; implemented VGA line-drawing algorithms in **Verilog** for FPGA deployment

Radar Systems Fundamentals — *Jan. 2024 – Mar. 2024*
Radar Research Lab, SINES | MATLAB — *Islamabad, Pakistan*

- Studied radar system fundamentals, focusing on beamforming and signal processing techniques using **MATLAB**, building foundational knowledge through MIT OpenCourseWare

Skills

Programming: Python, MATLAB, C/C++, Verilog, SQL, Bash

Machine Learning: PyTorch, TensorFlow, scikit-learn, GANs, VAEs, Deep RL, Generative AI, RAG, LangChain

Data & MLOps: PySpark, FastAPI, Snowflake, PostgreSQL, ETL Pipelines, Statistical Profiling, Feature Engineering, Synthetic Data Generation, MLflow, Optuna, Docker, Kubernetes, AWS

IC Design Tools: Cadence Virtuoso (ADE), Synopsys Custom Compiler (PrimeSim/PrimeWave)

Other Tools: LaTeX, Git, FPGA Design (Verilog)

Awards & Honors

NUST Merit Scholarship – Perfect 4.00 CGPA in multiple semesters — *2022 & 2023*

PM's Youth Laptop Scheme (Phase III) – Government of Pakistan — *2023*

Merit Scholarship – Pre-Engineering | Awarded for 2 consecutive years — *2018 – 2020*